

Beginner's Guide to Google Colab : Step by Step Tutorial

[01526102] Data Structure and Algorithm
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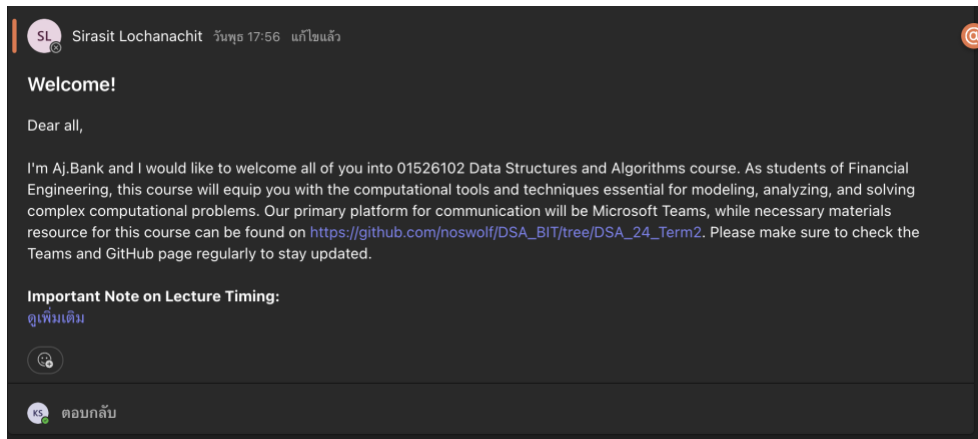
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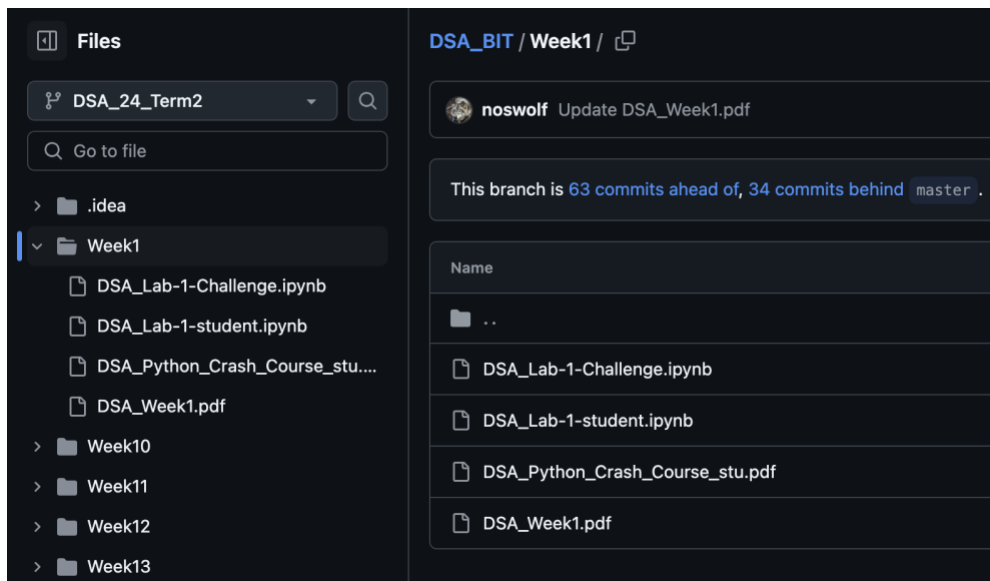
Beginner's Guide to Google Colab: Step by Step Tutorial

colab.google

The necessary materials resource for this course can be found on GitHub.

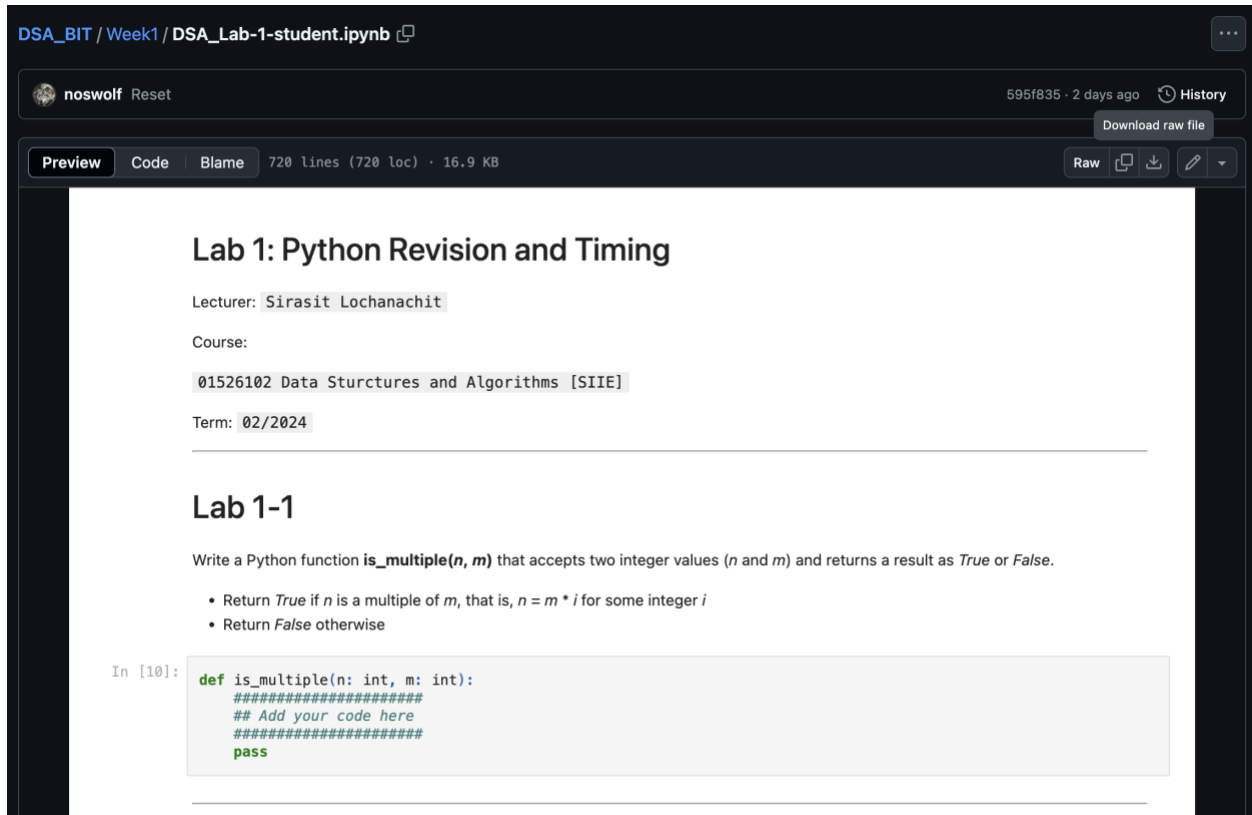


[GitHub - noswolf/DSA_BIT](https://github.com/noswolf/DSA_BIT) at DSA_24_Term2



Downloading Jupyter Notebooks from GitHub

Click the Download button in the top-right corner of the GitHub page to save the Jupyter Notebook.



The screenshot shows a Jupyter Notebook interface with a dark theme. At the top, the file path is `DSA_BIT / Week1 / DSA_Lab-1-student.ipynb`. Below the path bar, there's a header with the username `noswolf`, a `Reset` button, and a commit hash `595f835` with the text `2 days ago` and a `History` button. A `Download raw file` button is also present. Below the header, there's a tab bar with `Preview` (selected), `Code`, and `Blame`. The `Preview` tab shows the notebook content. The notebook has 720 lines (720 loc) and is 16.9 KB. The content includes a title `Lab 1: Python Revision and Timing`, a lecturer name `Sirasit Lochanachit`, a course name `01526102 Data Structures and Algorithms [SIIE]`, and a term `02/2024`. Below this, there's a section titled `Lab 1-1` with a description: `Write a Python function is_multiple(n, m) that accepts two integer values (n and m) and returns a result as True or False.` This is followed by a list of requirements:

- Return `True` if `n` is a multiple of `m`, that is, `n = m * i` for some integer `i`
- Return `False` otherwise

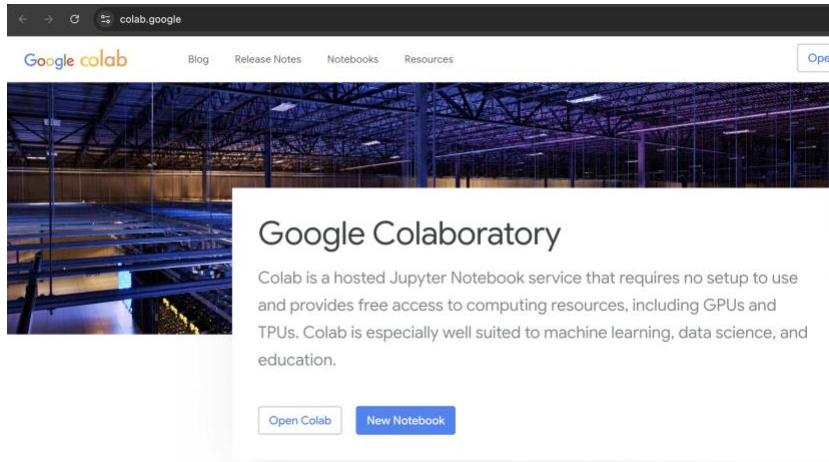
 At the bottom, there's a code cell with the prompt `In [10]:` and the following Python code:

```
def is_multiple(n: int, m: int):  
    #####  
    ## Add your code here  
    #####  
    pass
```

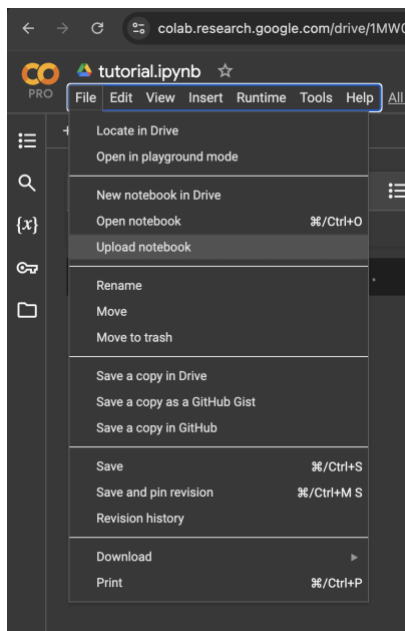
1. How to Copy and Use .ipynb Files in Your Environment

Uploading from Your Computer

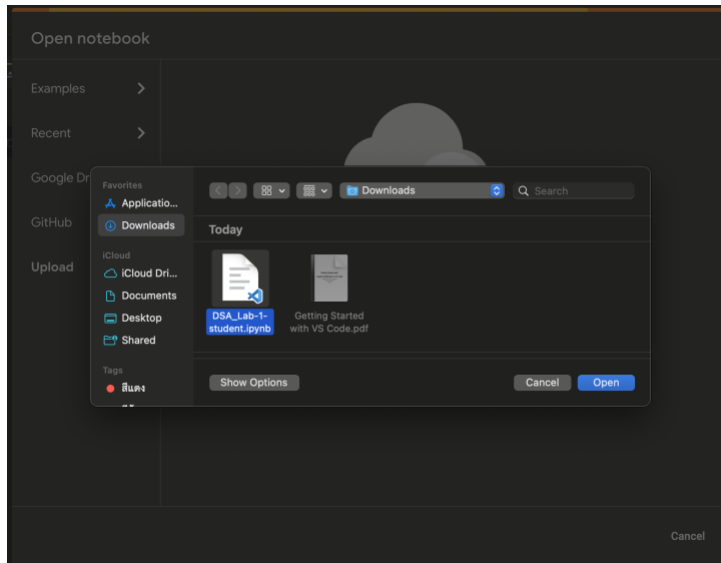
1. Go to [Google Colab](https://colab.google)
 1. Click "New Notebook"



2. Click "File" > "Upload notebook"



3. Pick your .ipynb file from your computer

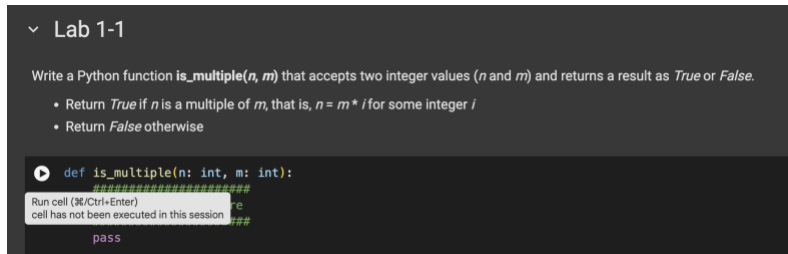


4. Done! You can now use the file

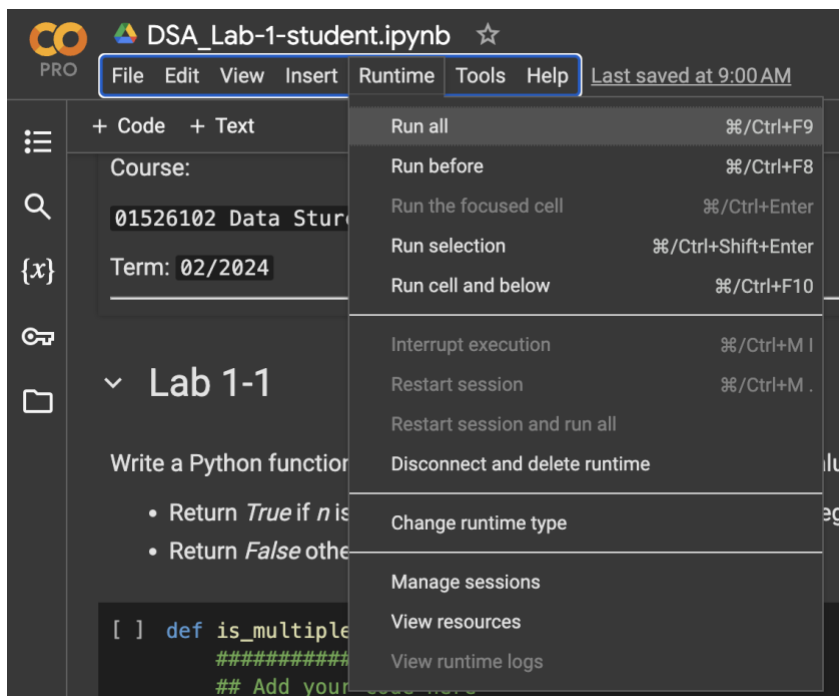
2. How to Run Code (The Basic Way)

Easy Ways to Run Your Code

1. Click the play button () next to your code

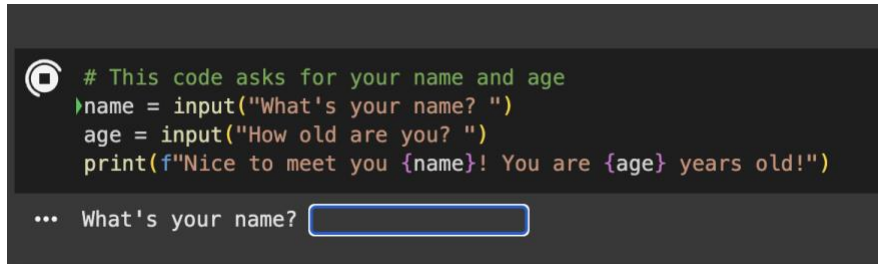


2. Or use Shift + Enter to run and move to the next cell
3. Or use Ctrl + Enter to run just the current cell (Mac : Command + Enter)
4. Or use run all and other options.



3. How to Run Code That Needs User Input

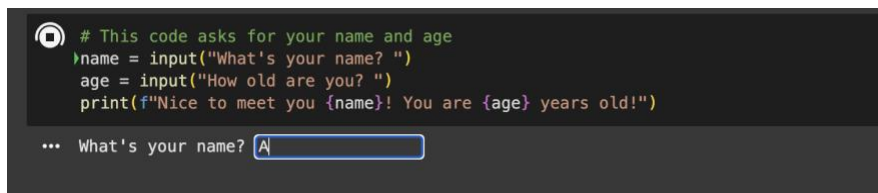
Simple Way: Using input()



```
# This code asks for your name and age
name = input("What's your name? ")
age = input("How old are you? ")
print(f"Nice to meet you {name}! You are {age} years old!")
```

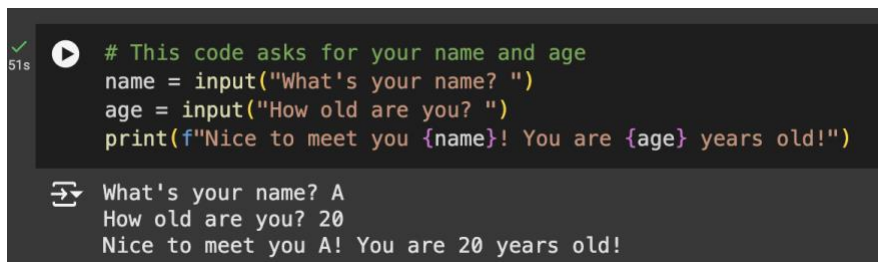
... What's your name?

Type input into the input box.



```
# This code asks for your name and age
name = input("What's your name? ")
age = input("How old are you? ")
print(f"Nice to meet you {name}! You are {age} years old!")
```

... What's your name?



```
# This code asks for your name and age
name = input("What's your name? ")
age = input("How old are you? ")
print(f"Nice to meet you {name}! You are {age} years old!")
```

What's your name? A
How old are you? 20
Nice to meet you A! You are 20 years old!

4. How to Install and Remove Libraries (Packages)

1. Installing New Package

Use “!pip install [package name]”

```
✓ 6s # To install one package
!pip install pandas

# To install multiple packages at once
!pip install matplotlib numpy seaborn

Requirement already satisfied: pandas in /usr/local/lib/python3.10/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.22.4 in /usr/local/lib/python3.10/dist-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.10/dist-packages (from pandas) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-packages (from pandas) (2024.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.10/dist-packages (from pandas) (2024.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.10/dist-packages (3.8.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.10/dist-packages (1.26.4)
Requirement already satisfied: seaborn in /usr/local/lib/python3.10/dist-packages (0.13.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (1.3.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (4.55.0)
Requirement already satisfied: kiwisolver>=1.0.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (1.4.7)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=6.2.0 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (11.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.10/dist-packages (from matplotlib) (2.8.2)
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.10/dist-packages (from seaborn) (2.2.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-packages (from pandas>=1.2->seaborn) (2024.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.10/dist-packages (from pandas>=1.2->seaborn) (2024.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
```

2. Checking What Version You Have

Use “!pip show [package name]”

```
✓ 3s # See what version you have
!pip show pandas

Name: pandas
Version: 2.2.2
Summary: Powerful data structures for data analysis, time series, and statistics
Home-page: https://pandas.pydata.org
Author:
Author-email: The Pandas Development Team <pandas-dev@python.org>
License: BSD 3-Clause License
```

3. Removing Packages You Don't Want

Use “!pip uninstall [package name] -y”

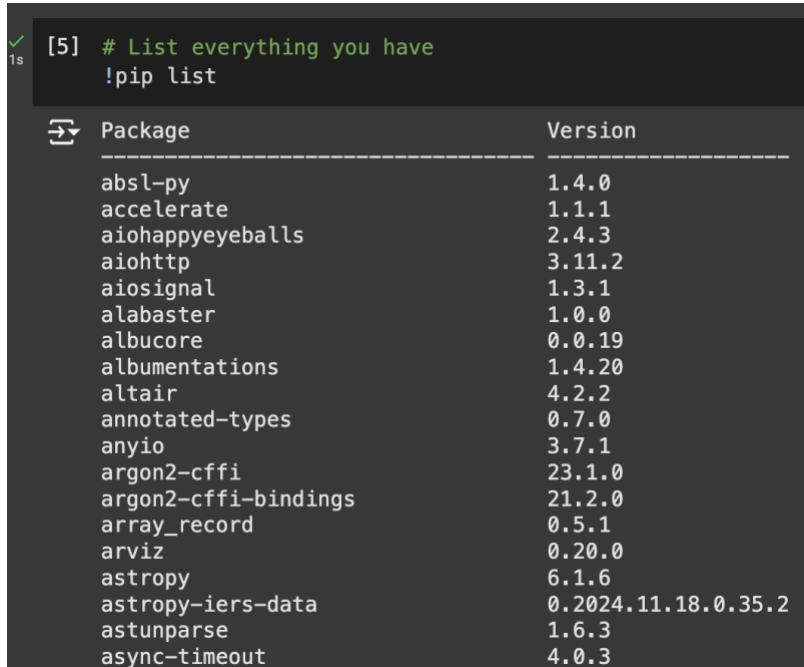
```
✓ 2s # Remove a package
!pip uninstall pandas -y

Found existing installation: pandas 2.2.2
Uninstalling pandas-2.2.2:
Successfully uninstalled pandas-2.2.2
```

Important Tips About Packages:

- After installing or removing packages:
 1. Click "Runtime" in the menu
 2. Click "Restart session"
 3. Click "Yes" when asked
 4. This makes sure your changes work properly

See All Installed Packages



```
[5] # List everything you have
!pip list
```

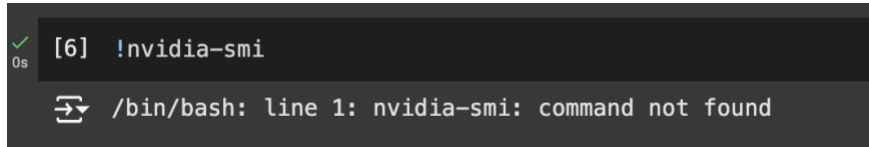
Package	Version
absl-py	1.4.0
accelerate	1.1.1
aiohappyeyeballs	2.4.3
aiohttp	3.11.2
aiosignal	1.3.1
alabaster	1.0.0
albucore	0.0.19
albumentations	1.4.20
altair	4.2.2
annotated-types	0.7.0
anyio	3.7.1
argon2-cffi	23.1.0
argon2-cffi-bindings	21.2.0
array_record	0.5.1
arviz	0.20.0
astropy	6.1.6
astropy-iers-data	0.2024.11.18.0.35.2
astunparse	1.6.3
async-timeout	4.0.3

Helpful Tips and Tricks

1. When Things Get Stuck

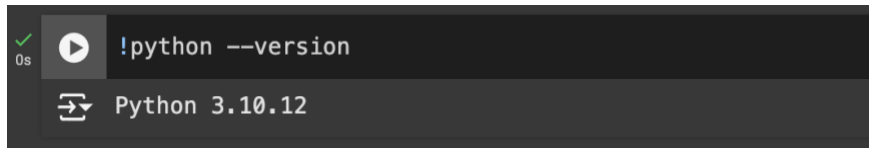
- Just restart! Click "Runtime" > "Restart session"
- Think of it like rebooting your phone when it's acting weird

2. Check if You Have GPU Power



```
[6] !nvidia-smi
/bin/bash: line 1: nvidia-smi: command not found
```

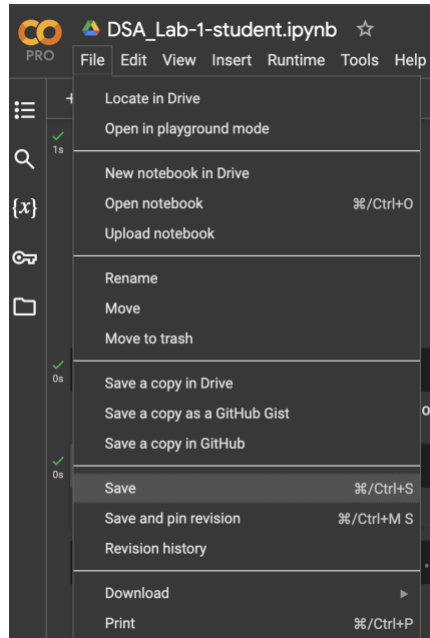
3. Check Your Python Version



```
!python --version
Python 3.10.12
```

4. Saving Your Work

- Press `Ctrl + S`, or
- Click "File" > "Save"



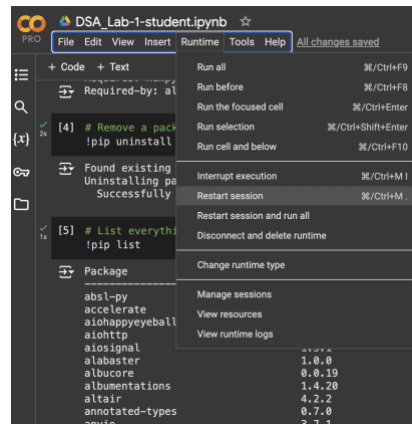
- Don't worry too much - Colab auto-saves every 2 minutes!

5. Cool Keyboard Shortcuts

- Press `Tab` while typing to auto-complete
- Press `Shift + Tab` inside a function to see what it does

6. Memory Problems?

- If your code is running slow or using too much memory:
 1. Go to "Runtime" > "Restart session"



2. This gives you a fresh start

7. Working with Files

- Your files don't stay in Colab forever
- Save important stuff to Google Drive
- Download anything you want to keep

8. Need Help?

- Click the "Help" menu